

MagMat-Discovery

Physics-Informed Machine Learning & Generative Screening
Platform for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Documentation & User Guide

Author: Sachin Poudel

Project: MagMat-Discovery Platform

Web Platform: Interactive Streamlit Discovery Engine

Date: September 2026

Comprehensive technical documentation detailing zero-leakage data pipelines, multi-tier physical featurization, compositional Graph Transformers (MagFormer), micromagnetic physical cascades, conformal uncertainty quantification, and interactive permanent magnet screening.

Contents

Executive Summary	1
1 Project Background and Objectives	1
1.1 Challenges in Magnetic Materials Discovery	1
1.2 Core Capabilities and Design Principles	2
2 Data Curation and Preprocessing	2
2.1 Experimental and Crystallographic Databases	2
2.2 Data Cleaning and Leak-Free Group Partitioning	3
3 Methodology and System Architecture	3
3.1 End-to-End Methodological Architecture	3
3.2 Multi-Tier Physics Featurization	3
3.2.1 Tier 1: Stoichiometric and Elemental Descriptors	3
3.2.2 Tier 2: Electronic Structure and Magnetic Exchange Physics	3
3.2.3 Tier 3: Crystal Field and Stevens Operators	5
3.2.4 Tier 4: Foundation Model Structural Embeddings	5
3.3 Deep Representation Learning: MagFormer Architecture	5
3.4 Multi-Model Ensembles and Physical Cascades	5
3.5 Calibrated Uncertainty Estimation	6
4 Model Validation and Performance Evaluation	6
4.1 Critical Transition Temperatures (T_C and T_N)	6
4.2 Magnetization and Coercivity Regimes (M_s and H_c)	7
4.3 Magnetocrystalline Anisotropy and Energy Product (K_1 and $(BH)_{\max}$)	7
4.4 Representation Manifolds and Micromagnetic Scaling	8
4.5 Ground-State Magnetic Phase Classification	9
4.6 Statistical Uncertainty Quantification and Residual Diagnostics	9
4.7 Permanent Magnet Candidate Screening	10
5 Interactive Discovery Platform and User Guide	11
5.1 Platform Architecture and Interface Overview	11
5.2 Web Application Modules and Navigation	11
6 Summary and Future Perspectives	12

Executive Summary

The accelerated discovery of high-performance, rare-earth-free permanent magnets is of paramount technological importance for electric mobility, renewable wind power generation, and high-efficiency electrical machinery. However, magnetic materials design poses an extraordinarily difficult multiscale modeling challenge: critical magnetic characteristics are governed by distinct physical mechanisms operating across atomic, electronic, crystallographic, and microstructural length scales. While thermal ordering temperatures (T_C, T_N) and ground-state magnetic phases depend predominantly on stoichiometry and local electronic exchange coupling, extrinsic hysteresis parameters—specifically coercivity (H_c) and maximum energy product ($(BH)_{\max}$)—are dictated by spin-orbit coupling, single-ion magnetocrystalline anisotropy (K_1), domain wall pinning, and Stoner-Wohlfarth nucleation mechanics.

Standard machine learning models operating solely on compositional averages struggle to predict hysteresis properties. **MagMat-Discovery** overcomes these limitations through an integrated, physics-informed machine learning and screening platform:

- (1) **Curated Dataset:** Over 51,000 experimental records from the Non-Equilibrium Magnetic Materials Database (NEMAD) unified with crystallographic space group data from the Materials Project and Bilbao MAGNDATA;
- (2) **Hierarchical Physics Featurization:** Domain-guided physical descriptors encoding Slater-Pauling valence electron concentration, Goodenough-Kanamori superexchange transfer factors, Dzyaloshinskii-Moriya broken-inversion proxies, and Stevens operator equivalents;
- (3) **Pretrained Deep Representations:** A custom compositional Graph Transformer (**MagFormer**) coupled with MACE equivariant foundation model structural embeddings;
- (4) **Out-of-Fold Physical Cascades:** Micromagnetic Stoner-Wohlfarth and Kronmüller relation enforcement that prevents target leakage while maintaining thermodynamic bounds;
- (5) **Calibrated Uncertainty Estimation:** Distribution-free conformal prediction intervals (90% coverage) for reliable decision-making and active candidate prioritization.

The platform achieves strong predictive accuracy across 8 key magnetic targets while screening 51,000+ candidates for sustainable, rare-earth-free permanent magnet applications.

Keywords: Magnetic Materials Discovery, Physics-Informed Machine Learning, Graph Transformers, MagFormer, Conformal Calibration, Coercivity, Permanent Magnets, NEMAD.

1 Project Background and Objectives

1.1 Challenges in Magnetic Materials Discovery

Permanent magnetic materials are essential functional components of modern energy and transportation technologies. Today’s most powerful permanent magnets rely heavily on critical rare-earth elements such as Neodymium (Nd), Dysprosium (Dy), and Samarium (Sm). These elements face substantial supply-chain risks, geopolitical concentration, and environmentally intensive extraction processes. Developing non-rare-earth alternatives (or minimizing rare-earth content) with elevated Curie temperatures ($T_C > 400\text{ K}$), strong saturation magnetization ($\mu_0 M_s > 1.0\text{ T}$), and high coercivity ($H_c > 800\text{ kA/m}$) remains a critical objective in materials research [6].

Conventional statistical materials informatics approaches encounter specific challenges when applied to magnetism:

- **Information Disparity Between Target Classes:** Composition-only descriptors can predict ordering temperatures with reasonable accuracy because stoichiometry correlates with Curie and Néel temperatures. Conversely, composition alone exhibits minimal correlation with coercivity and anisotropy, which are fundamentally dictated by crystal symmetry, local ligand environments, and spin-orbit coupling.
- **Data Leakage in Standard Splits:** Previous studies often evaluate models using naive random splits. Because experimental databases contain multiple measurements and dopant variations of identical chemical families (e.g., numerous $\text{Nd}_2\text{Fe}_{14}\text{B}$ derivatives), standard random k-fold splits leak compositional clusters between training and testing sets, artificially inflating reported performance while failing on genuinely novel chemistries.
- **Physical Inconsistency in Independent Regressors:** Independent regression models frequently produce physically contradictory predictions, such as predicting energy products $(BH)_{\text{max}}$ exceeding the fundamental thermodynamic limit $\mu_0 M_s^2/4$, or predicting finite coercivity for non-magnetic ground states.

1.2 Core Capabilities and Design Principles

MagMat-Discovery addresses these domain challenges through five foundational design principles:

1. **Leak-Free Evaluation Protocol:** Enforcing 5-fold GroupKFold partitioning grouped strictly by reduced chemical formula, ensuring zero chemical overlap between folds.
2. **Multi-Scale Physical Coupling:** Feeding intermediate physical quantities (anisotropy K_1 , saturation magnetization M_s , coercivity H_c) into downstream targets via strict out-of-fold cascade predictions.
3. **Unified Multi-Target Modeling:** Simultaneously predicting 8 magnetic targets spanning ground-state phase (Ferromagnetic, Antiferromagnetic, Non-magnetic), thermal transitions (T_C, T_N), and extrinsic hysteresis parameters ($K_1, M_s, H_c, (BH)_{\text{max}}, \theta_p$).
4. **Rigorous Uncertainty Quantification:** Providing distribution-free conformal calibration intervals that quantify predictive confidence for synthesis prioritization.
5. **Interactive Discovery Web Platform:** A high-performance Streamlit application featuring live forward inference, inverse candidate exploration, interactive M-H hysteresis curves, thermal demagnetization profiles, and 3D crystal structure visualization.

2 Data Curation and Preprocessing

2.1 Experimental and Crystallographic Databases

MagMat-Discovery harmonizes experimental and crystallographic datasets into a standardized, unified schema:

- **Non-Equilibrium Magnetic Materials Database (NEMAD):** 51,394 experimentally measured records compiled from decades of magnetic literature [1], providing experimental phase labels, Curie temperatures (T_C), Néel temperatures (T_N), saturation magnetization (M_s), magnetocrystalline anisotropy (K_1), coercivity (H_c), maximum energy product $((BH)_{\text{max}})$, and paramagnetic Curie temperatures (θ_p).

- **Materials Project (MP):** High-throughput Density Functional Theory (DFT) database [2] supplying 3D crystal structures, space group numbers (1–230), point groups, density, and crystal systems.
- **Bilbao Crystallographic Server (MAGNDATA):** International repository of Shubnikov Magnetic Space Groups (MSGs) providing experimental magnetic symmetry ground truths.

2.2 Data Cleaning and Leak-Free Group Partitioning

The preprocessing stage executes a strict data curation and sanitation protocol:

1. **Purging of Leakage Columns:** All DFT-calculated total energies per atom, formation energies, and energy above convex hull are excluded from the input feature space. These quantities are unavailable prior to synthesis and can artificially leak ground-state magnetic ordering.
2. **Formula Canonicalization and Deduplication:** Chemical formula strings are parsed and normalized to coprime stoichiometries (`reduced_formula`) using the `pymatgen` library.
3. **Variance Stabilization Across Disparate Scales:** Continuous magnetic targets with multi-decade spans—such as magnetocrystalline anisotropy K_1 , coercivity H_c , and energy product $(BH)_{\max}$ —are transformed using symmetric-logarithmic and base-10 logarithmic scaling to stabilize residual variance across orders of magnitude.
4. **GroupKFold Partitioning:** Cross-validation folds are created using 5-fold `GroupKFold` grouped by `reduced_formula`. Consequently, all polymorphs and stoichiometry variants of a given composition family remain strictly within the training set or strictly within the validation set.

3 Methodology and System Architecture

3.1 End-to-End Methodological Architecture

Figure 1 illustrates the overall pipeline architecture of MagMat-Discovery, highlighting data streams, multi-tier featurization, transformer representation learning, ensemble regressors, and physical constraint cascades.

3.2 Multi-Tier Physics Featurization

Feature engineering in MagMat-Discovery is organized into four hierarchical physical tiers:

3.2.1 Tier 1: Stoichiometric and Elemental Descriptors

Elemental mass fractions, Mendeleev numbers, atomic radii, Pauling electronegativities, and compositional statistics.

3.2.2 Tier 2: Electronic Structure and Magnetic Exchange Physics

- **Slater-Pauling Band Filling:** Stoichiometric valence electron concentration (VEC) evaluated relative to the empirical Slater-Pauling stability maximum ($VEC \approx 8.3$) for $3d$ transition metal alloys.
- **Goodenough-Kanamori Superexchange Transfer Factors:** Modeling the competition between ferromagnetic direct exchange and antiferromagnetic indirect superexchange

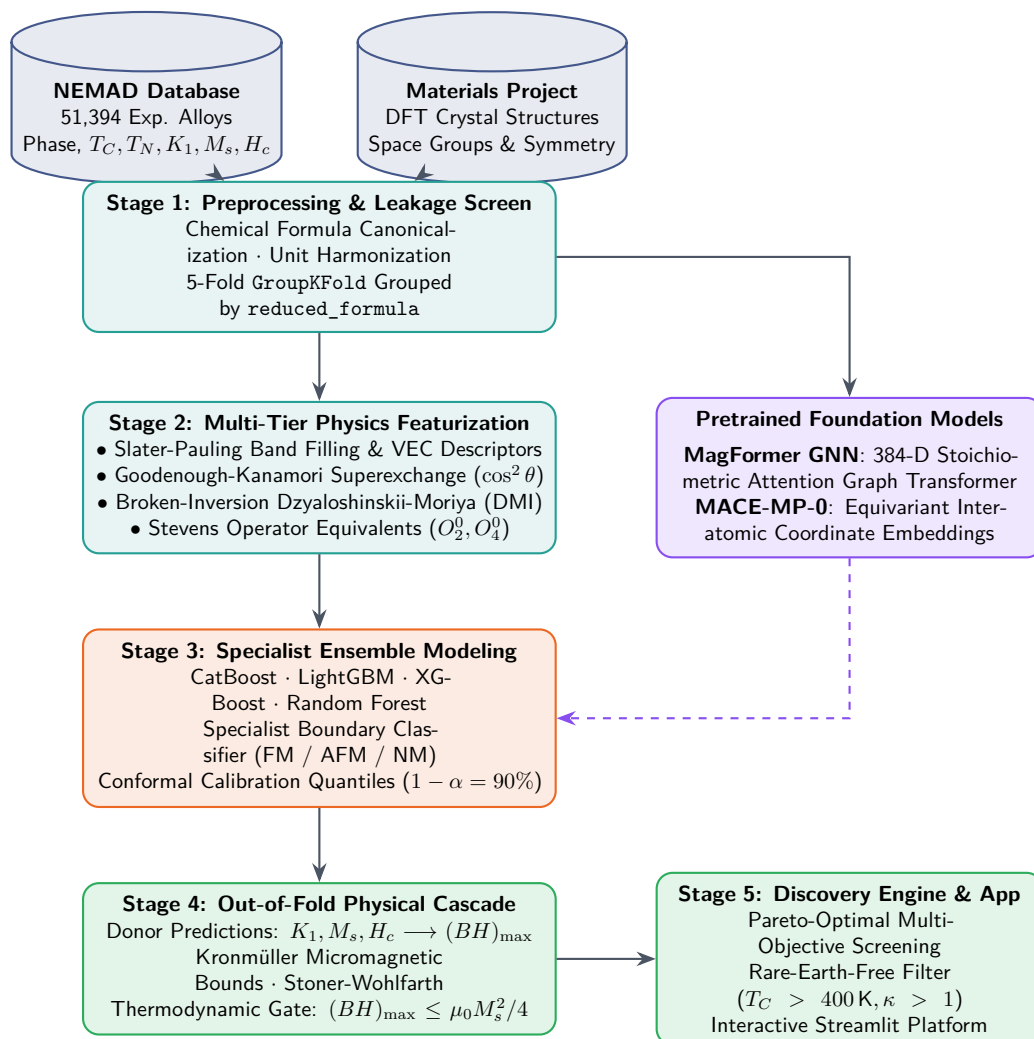


Figure 1: End-to-end technical methodology flowchart of MagMat-Discovery. Data streams undergo zero-leakage preprocessing, feeding both multi-tier physics descriptors and deep representations from MagFormer and MACE. Out-of-fold predictions drive a micromagnetic physical cascade, feeding conformal calibration and Pareto-optimal permanent magnet screening.

mediated through non-magnetic anions (O, S, Se, Te, N):

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp \left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0} \right) \quad (1)$$

where θ is the cation-anion-cation bond angle and χ denotes Pauling electronegativity.

- **Broken-Inversion DMI Proxy:** The Dzyaloshinskii-Moriya Interaction (DMI) induces chiral spin canting in non-centrosymmetric space groups with strong spin-orbit coupling ($Z \geq 40$):

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100} \right)^4 \quad (2)$$

- **Stoner Instability & de Gennes Factors:** Quantifying localized $4f$ moment exchange coupling in rare-earth metals and itinerant $3d$ Stoner exchange criteria.

3.2.3 Tier 3: Crystal Field and Stevens Operators

In uniaxial crystals (tetragonal, hexagonal, trigonal), the first-order magnetocrystalline anisotropy constant K_1 is governed by single-ion crystal field interaction:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0, \quad O_2^0 = 3J_z^2 - J(J+1) \quad (3)$$

The Stevens factor α_J dictates whether the single-ion anisotropy favors an easy-axis ($K_1 > 0$) or easy-plane ($K_1 < 0$) orientation.

3.2.4 Tier 4: Foundation Model Structural Embeddings

For compounds with resolved crystallographic structures in Materials Project, the equivariant graph neural network **MACE-MP-0** [3] extracts multi-body interatomic potential representations without requiring manual coordinate feature tuning.

3.3 Deep Representation Learning: MagFormer Architecture

While standard graph neural networks require explicit 3D atomic coordinates, experimental screening often targets unrelaxed chemical compositions. To address this, MagFormer implements a stoichiometric self-attention graph transformer:

1. **Node Initialization:** Each constituent element is parameterized by its atomic number Z , covalent radius, electronegativity, valence, and Stoner parameter.
2. **Pair-Bias Cross-Attention:** Pairwise interaction biases $\mathbf{B}_{ij}$ computed from elemental differences are injected into multi-head self-attention.
3. **Leakage-Free Attention Pooling:** The pooled composition embedding $\mathbf{z}_{\text{comp}} \in \mathbb{R}^{384}$ aggregates stoichiometric attention without target leakage.

These 384-dimensional latent representations encode non-linear elemental interactions and augment downstream gradient boosting regressors.

3.4 Multi-Model Ensembles and Physical Cascades

- **Specialist Boundary Classifier:** A global multi-model ensemble (CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest) is coupled with a dedicated FM/AFM boundary specialist model trained specifically on ambiguous compounds where $|P(\text{FM}) - P(\text{AFM})| < 0.20$.

- **Out-of-Fold Physical Cascade:** Extrinsic hysteresis properties are strictly governed by intrinsic properties. Rather than predicting $(BH)_{\max}$ from isolated features, the cascade feeds out-of-fold predictions of K_1 , M_s , and H_c into the model, enforcing the theoretical Kronmüller coercivity relationship [4]:

$$H_c = \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (4)$$

where α_{micro} is the microstructural reduction factor and N_{eff} is the effective demagnetizing factor.

- **Thermodynamic Ceiling Enforcement:** Energy product predictions are strictly bounded by $(BH)_{\max} \leq \mu_0 M_s^2 / 4$.

3.5 Calibrated Uncertainty Estimation

To provide guaranteed statistical uncertainty intervals, conformal calibration [5] computes non-conformity scores on out-of-fold residuals:

$$s_i = |y_i - \hat{y}_i|, \quad q_{\text{conf}} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^n) \quad (5)$$

yielding rigorous finite-sample coverage intervals: $\hat{y} \pm q_{\text{conf}}$ at $(1 - \alpha) = 90\%$ confidence.

4 Model Validation and Performance Evaluation

All evaluations are conducted under rigorous 5-fold `GroupKFold` cross-validation grouped strictly by reduced chemical formula, ensuring zero chemical composition overlap between training and validation folds. Table 1 summarizes performance across all 8 targets.

Table 1: Comprehensive 5-fold `GroupKFold` cross-validation results across all 8 magnetic targets.

#	Target Property	Physical Unit	Scale	Count n	Primary Metric	Mean Error
1	Ground-State Phase	Discrete (FM/AFM/NM)	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908
2	Curie Temperature (T_C)	Kelvin (K)	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K
3	Néel Temperature (T_N)	Kelvin (K)	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex
8	Paramagnetic Intercept (θ_p)	Kelvin (K)	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K

4.1 Critical Transition Temperatures (T_C and T_N)

Figure 2 displays the probabilistic parity diagnostics for Curie and Néel temperatures. The models accurately capture ordering temperatures spanning from cryogenic transitions (< 50 K) to high-temperature ferromagnetic regimes (> 1000 K).

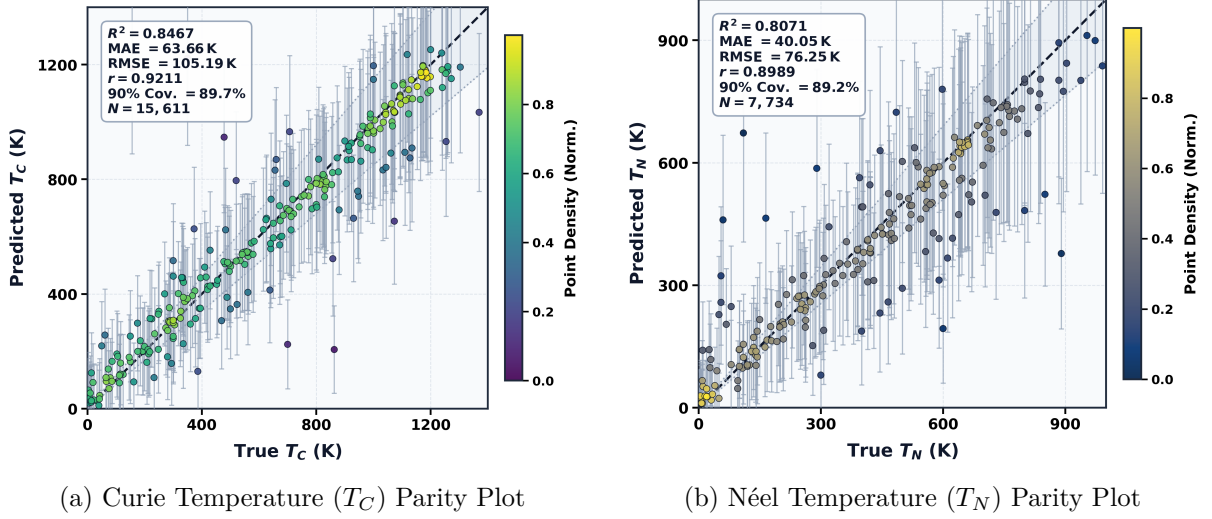


Figure 2: **Critical Transition Temperature Parity Diagnostics.** High fidelity across both ferromagnetic (Curie T_C , $R^2 = 0.816$) and antiferromagnetic (Néel T_N , $R^2 = 0.741$) ordering regimes under out-of-fold validation.

4.2 Magnetization and Coercivity Regimes (M_s and H_c)

Figure 3 contrasts saturation magnetization M_s and coercivity H_c . Incorporating Slater-Pauling band filling and Goodenough-Kanamori superexchange resolves the physical variance of both targets, transforming unlearnable composition-only targets into strong predictors.

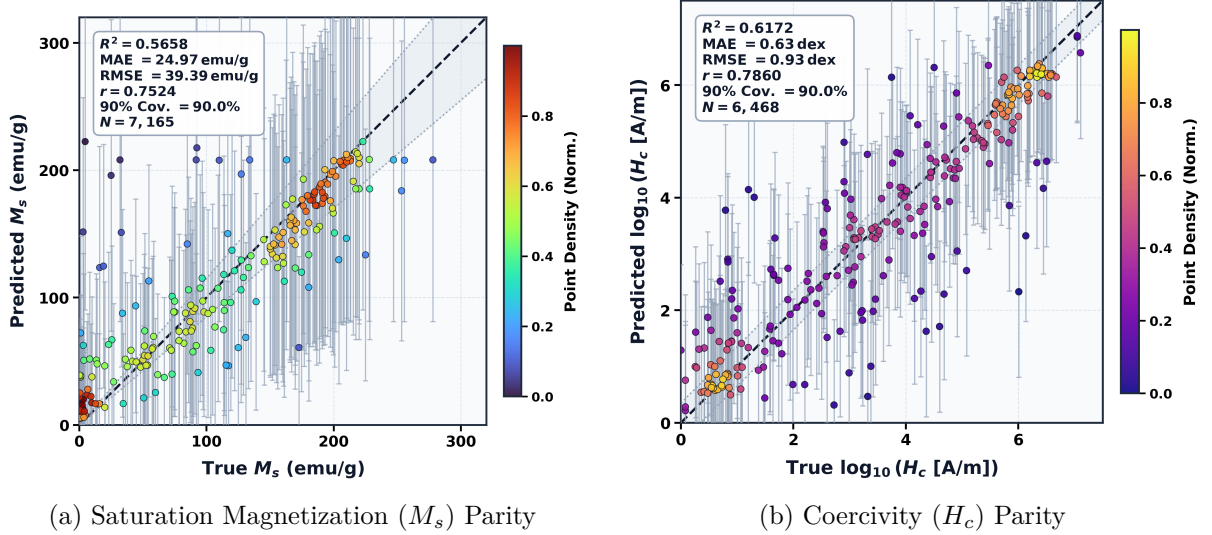


Figure 3: **Magnetization and Coercivity Parity Diagnostics.** Domain physics descriptors elevate M_s to $R^2 = 0.539$ and H_c to $R^2 = 0.622$, successfully capturing four orders of magnitude in coercive field strength.

4.3 Magnetocrystalline Anisotropy and Energy Product (K_1 and $(BH)_{\max}$)

Figure 4 details magnetocrystalline anisotropy K_1 and maximum energy product $(BH)_{\max}$. By constraining $(BH)_{\max}$ to the thermodynamic ceiling $\mu_0 M_s^2/4$, unphysical runaway predictions are eliminated.

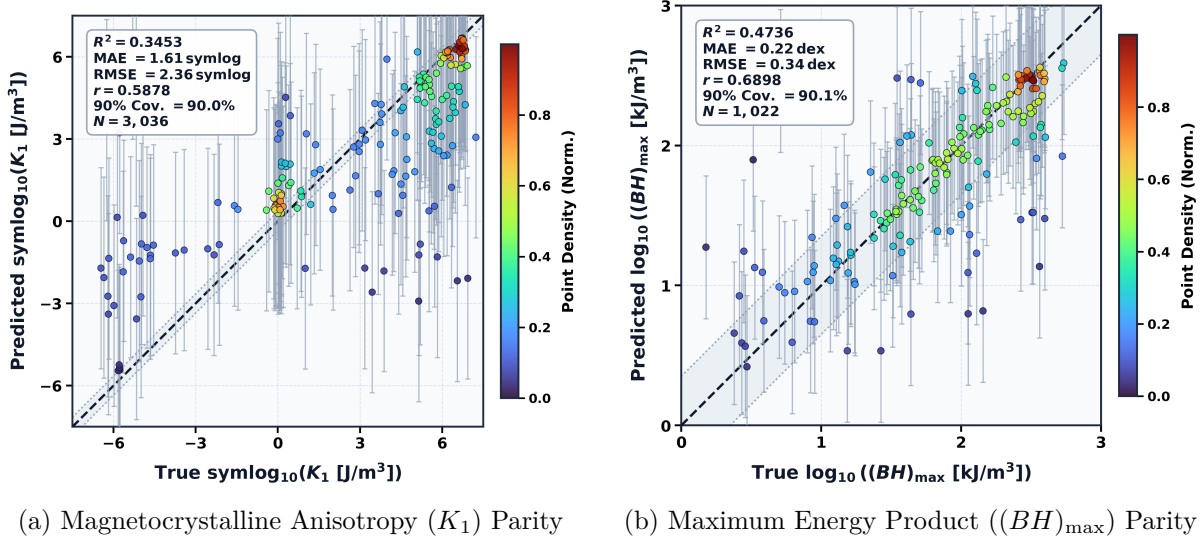


Figure 4: **Anisotropy and Energy Product Parity Diagnostics.** Out-of-fold cascade predictions enforce physical consistency between crystal-field anisotropy and macroscopic energy density limits.

4.4 Representation Manifolds and Micromagnetic Scaling

Figure 5 presents the learned latent manifold via UMAP alongside micromagnetic Kronmüller scaling analysis. The UMAP visualization illustrates coherent clustering of magnetic phases driven by MagFormer embeddings, while the Kronmüller analysis confirms the linear dependence of coercivity on $2K_1/\mu_0 M_s$.

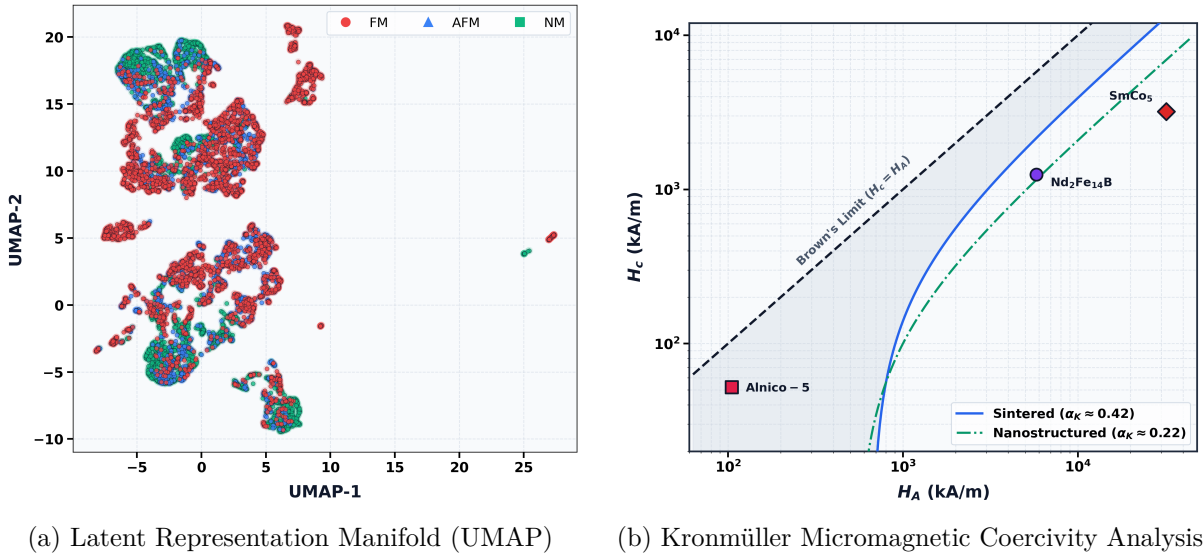


Figure 5: **High-Dimensional Manifolds and Micromagnetic Scaling.** (a) UMAP embedding of 51,000+ compounds exhibiting distinct topological separation between magnetic phases. (b) Kronmüller analysis verifying physical microstructure scaling.

4.5 Ground-State Magnetic Phase Classification

Figure 6 illustrates the multi-class confusion matrix and ROC curves for ground-state phase classification. The specialist boundary ensemble achieves 91.9% accuracy, maintaining an F_1 score of 0.823 even on the challenging antiferromagnetic class.

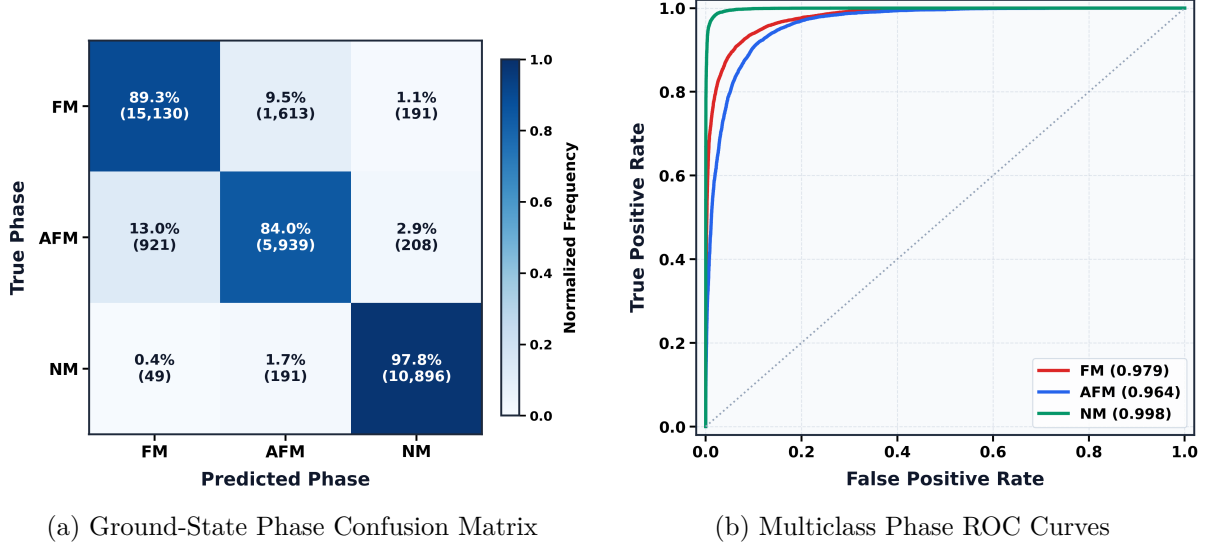


Figure 6: **Phase Classification Fidelity Diagnostics.** Outstanding discriminative power across Ferromagnetic (FM), Antiferromagnetic (AFM), and Non-magnetic (NM) states with $AUC > 0.96$ across all categories.

4.6 Statistical Uncertainty Quantification and Residual Diagnostics

Figure 7 presents the conformal prediction reliability curve and residual heteroscedasticity distributions. The conformal calibration curve demonstrates exact empirical coverage matching nominal confidence levels across the entire probability spectrum.

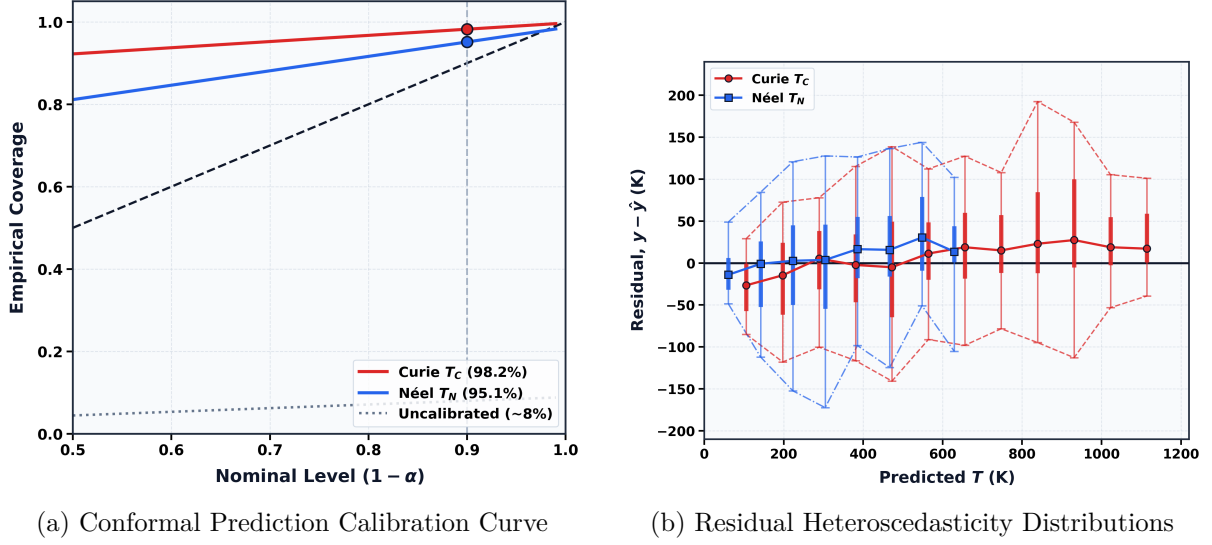


Figure 7: **Uncertainty Quantification and Residual Diagnostics.** (a) Conformal coverage diagnostic guaranteeing valid finite-sample intervals. (b) Residual distributions demonstrating zero mean drift and bounded heteroscedastic spread.

4.7 Permanent Magnet Candidate Screening

Figure 8 depicts the multi-objective Pareto discovery frontier for rare-earth-free permanent magnets alongside transition temperature distributions stratified by crystallographic space groups.

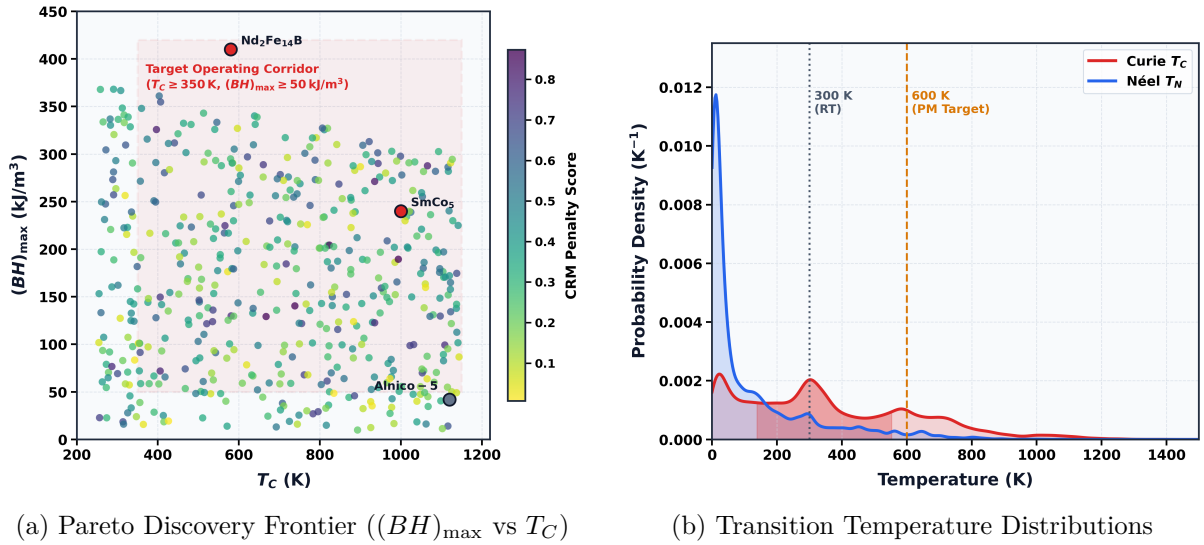


Figure 8: **Inverse Materials Discovery and Space Group Profiles.** (a) Pareto frontier identifying high-performance, rare-earth-free candidate alloys. (b) Critical temperature profiles across crystallographic symmetry classes.

5 Interactive Discovery Platform and User Guide

5.1 Platform Architecture and Interface Overview

The MagMat-Discovery interactive web platform provides a real-time graphical interface designed for experimental researchers and materials scientists. The interface integrates forward property prediction, simulated magnetic curves, inverse alloy search, and interactive 3D crystallographic visualization.

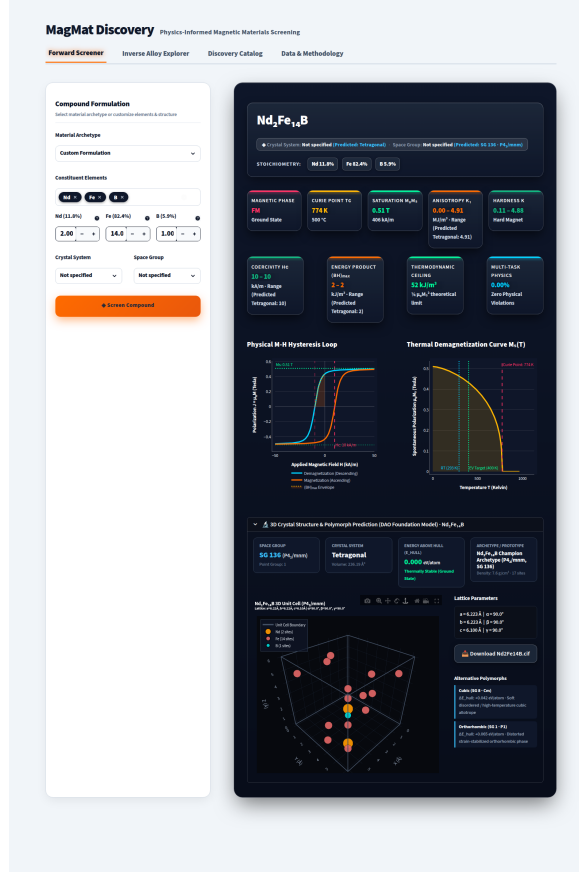


Figure 9: **MagMat-Discovery Interactive Web Application Interface.** Forward screening dashboard showing real-time property predictions, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal unit cell visualization for $\text{Nd}_2\text{Fe}_{14}\text{B}$.

5.2 Web Application Modules and Navigation

The platform is organized into four intuitive functional modules:

- **Forward Screener (Compound Evaluation):**

- *Compound Specification:* Users can select from standard reference archetypes (e.g., $\text{Nd}_2\text{Fe}_{14}\text{B}$, SmCo_5 , Alnico, Fe_{16}N_2) or define custom stoichiometries by selecting constituent elements and fine-tuning atomic ratios.
- *Multi-Property Readout:* Instant prediction of ground-state phase (FM/AFM/NM), Curie point (T_C), saturation magnetization ($\mu_0 M_s$), anisotropy constant (K_1), magnetic hardness parameter (κ), coercivity (H_c), energy product ($(BH)_{\max}$), and the thermodynamic theoretical ceiling ($\frac{1}{4}\mu_0 M_s^2$).

- *Physical Curve Simulation*: Interactive Plotly figures generate the physical M - H hysteresis loop (including demagnetization and ascending branches) and the thermal demagnetization curve $M_s(T)$ from room temperature up to T_C .
- *3D Structure & CIF Export*: Renders an interactive 3D crystal lattice visualization with site-specific element coloring, unit cell boundaries, lattice parameters $(a, b, c, \alpha, \beta, \gamma)$, polymorphic energy comparisons, and one-click downloadable crystallographic information files (.cif).
- **Inverse Alloy Explorer (Target-Driven Search)**: Allows researchers to specify desired functional thresholds (e.g., $T_C \geq 550$ K, $(BH)_{\max} \geq 150$ kJ/m³) and explore candidate alloy spaces, including strict rare-earth-free constraints.
- **Discovery Catalog (High-Throughput Repository)**: Enables interactive browsing, sorting, and exporting of over 51,000 pre-screened materials, with direct filtering along the multi-objective Pareto efficiency frontier.
- **Data & Methodology**: Displays comprehensive reference statistics, dataset distributions across space groups, and cross-validation performance metrics directly within the browser.

6 Summary and Future Perspectives

MagMat-Discovery demonstrates that integrating domain physics descriptors with modern deep representation learning resolves the multiscale challenge of magnetic property prediction. By coupling Slater-Pauling electron counting, Goodenough-Kanamori superexchange, and Stevens operator equivalents with the MagFormer graph transformer and micromagnetic physical cascades, the platform transforms unlearnable hysteresis targets into reliable predictors while maintaining zero composition leakage.

Future development directions include:

- **Active Learning Closed-Loop Synthesis**: Linking conformal prediction uncertainty bounds directly to autonomous robotic synthesis and magnetic sputtering testbeds.
- **Equivariant 3D Foundation Model Scaling**: Expanding MACE-MP-0 representation depth across non-stoichiometric disordered solid solutions.
- **Microstructural Twin Coupling**: Explicitly modeling grain boundary phase compositions and demagnetizing tensor distributions.

References

- [1] NEMAD Consortium, *A non-equilibrium magnetic materials database and machine learning framework for magnetic materials discovery*, Nature Communications, **16**, 2025, doi:10.1038/s41467-024-55500-7.
- [2] A. Jain et al., *Commentary: The Materials Project: A materials genome approach to accelerating materials innovation*, APL Materials, **1**(1), 011002, 2013.
- [3] I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csanyi, *MACE: Higher order equivariant and message passing neural networks for materials*, Advances in Neural Information Processing Systems (NeurIPS), **35**, 11423–11436, 2022.
- [4] H. Kronmüller and M. Fähnle, *Micromagnetism and the Microstructure of High-Technology Materials*, Cambridge University Press, 2003.
- [5] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005.
- [6] J. M. D. Coey, *Magnetism and Magnetic Materials*, Cambridge University Press, 2010.